Write your name here

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Master Functional Skills	Model Solutions (MFS)

Pearson Edexcel Functional Skills

Centre Number	Candidate Number
MFS-UK	L2-MAR18

Mathematics

Level 2



19 – 23 March 2018
Time: 1 hour 30 minutes

Paper Reference
FSM02/01

You must have:
Pen, calculator, HB pencil, eraser, ruler graduated in cm and mm, protractor, compasses.

Total Marks
48 /48

My signature confirms that I will not discuss the content of the test with anyone until the end of the 5 day test window.

Signature: Master Functional Skills (MFS)

Instructions

- Use a **black** ball-point pen.
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Sign the declaration.
- Answer **all** questions.
- Answer the questions in the spaces provided
  – there may be more space than you need.
- **Calculators may be used.**

Information

- The total mark for this paper is 48.
- The marks for each question are shown in brackets
  – use this as a guide to how much time to spend on each question.
- **You must show clearly how you get your answers because marks will be awarded for your working out.**
- **Check your working and your answers at each stage.**
- **This sign shows where marks will be awarded for showing your check.**

Advice

- Read each question carefully before you start to answer it.
- Keep an eye on the time.

Turn over ▶

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SECTION A: Fitness club

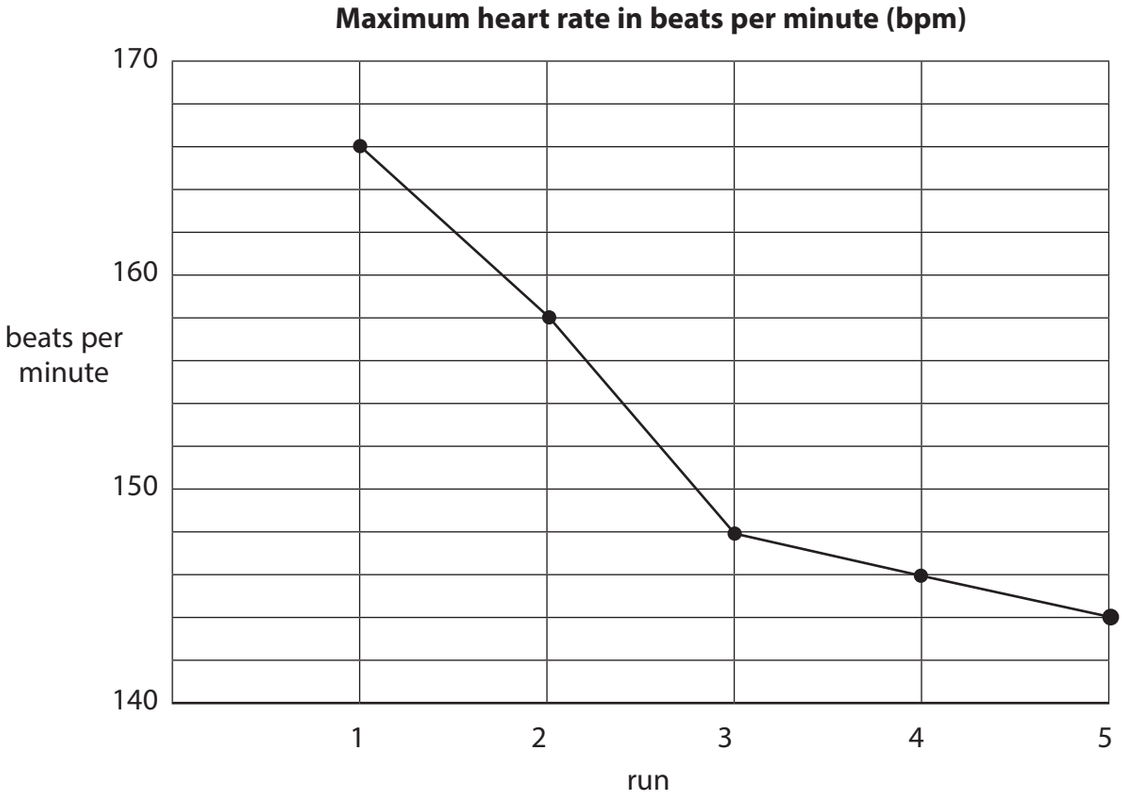
Answer all questions in this section.

Write your answers in the spaces provided.

1 Bradley is a member of a fitness club.

He plans to take part in a race.
Bradley does some training runs to prepare for the race.

He has this information about his maximum heart rate during each run.



Bradley thinks his mean maximum heart rate is greater than 150 beats per minute.

Is Bradley correct?
Show a check of your working. **(4)**

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Use the box below to show clearly how you get your answer.

Step 1: Read values from line graph and sum the 5 weeks [M1 readings]

Readings: W1 = 166, W2 = 158, W3 = 148, W4 = 146, W5 = 144 bpm

Total Heart Rate Sum = 166 + 158 + 148 + 146 + 144 = 762 bpm

Step 2: Calculate mean weekly heart rate (Sum ÷ 5) [M1 mean]

Mean = 762 ÷ 5 = 152.4 bpm

Step 3: Compare calculated mean with Bradley's claim (> 150 bpm)

152.4 > 150 (mean is 2.4 bpm above 150)

[A1 decision]

Conclusion: Yes, Bradley is correct (mean is 152.4 bpm > 150 bpm).

Use the box below to show your check.

☑

Reverse multiplication check: [B1 check]

152.4 × 5 = 762 bpm (matches total sum of readings)

Alternative check: 5 × 150 = 750; 762 > 750 (confirms mean > 150) (correct)

(Total for Question 1 is 4 marks)

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2 Bradley wants to buy a bike to help with his training.

He sees these two offers for the same make of bike.

Best Bikes	**Top Cycles**
Normal price £499	Normal price £369.59
Now $\frac{1}{8}$ off normal price	Plus VAT at 20%

Bradley wants to pay the least amount of money possible for the bike.

(a) Which offer should he choose?
Show why you think this. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate discounted price at Best Bikes (1/8 off £499) *[M1 best bikes]*

Discount = 1/8 × £499 = £499 ÷ 8 = £62.375

Best Bikes Price = £499 − £62.375 = £436.625 ≈ £436.63

Step 2: Calculate total price at Top Cycles (£369.59 + 20% VAT) *[M1 top cycles]*

VAT amount = 20% of £369.59 = 0.20 × £369.59 = £73.918

Top Cycles Price = £369.59 + £73.918 = £443.508 ≈ £443.51

(Alternative: 1.20 × £369.59 = £443.508)

Step 3: Compare prices (Bradley wants the cheapest offer)

£436.63 < £443.51 (Best Bikes is cheaper by £6.88)

[A1 decision]

Conclusion: Bradley should choose Best Bikes (Total £436.63).

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Bradley goes on a 25 km bike ride each week.

He wants to monitor his recovery heart rate.
Bradley records his recovery heart rate 10 minutes after each bike ride.

He has this information.

	Recovery heart rate (bpm)
ride 1	162
ride 7	135

Bradley thinks his recovery heart rate from ride 1 to ride 7 has decreased by $\frac{1}{5}$.

(b) Has his recovery heart rate decreased by $\frac{1}{5}$?

(3)

Use the box below to show clearly how you get your answer.

Step 1: Calculate actual decrease in heart rate *[M1 decrease]*

Start = 162 bpm, Finish = 135 bpm

Decrease = 162 − 135 = 27 bpm

Step 2: Calculate 1/5 of the starting heart rate (162 bpm) *[M1 fraction]*

1/5 of 162 = 162 ÷ 5 = 32.4 bpm

Step 3: Compare decrease with 1/5 threshold (32.4 bpm)

27 < 32.4 (the decrease was 5.4 bpm LESS than 1/5)

(Fraction decrease: 27/162 = 1/6 ≈ 16.7% < 20%)

[A1 decision]

Conclusion: No, Bradley is NOT correct (decreased by 27 bpm < 32.4 bpm).

(Total for Question 2 is 7 marks)

3 The probability of rain for the day of the race is 62%.

(a) What is the probability that it does **not** rain on the day of the race?

(2)

Use the box below to show clearly how you get your answer.

Probability that it does NOT rain on race day: *[M1 sub]*

Total Probability = 100% (or 1.0)

P(No Rain) = 100% − 62% = 38% (or 0.38)

[A1 prob]

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Runners who complete the race in 2 hours 15 minutes or less get a gold medal.

Bradley completed the race in 2 hours 13 minutes 22 seconds.

(b) Work out the difference between 2 hours 15 minutes and 2 hours 13 minutes 22 seconds. **(3)**

Use the box below to show clearly how you get your answer.

Step 1: Identify gold medal target time and Bradley's race time *[M1 times]*

Target Gold Medal Time = 2 hours 15 minutes 00 seconds

Bradley's Time = 2 hours 13 minutes 22 seconds

Step 2: Calculate the difference in time (subtraction / build-up method) *[M1 diff]*

From 2h 13m 22s to 2h 14m 00s = 60 − 22 = 38 seconds

From 2h 14m 00s to 2h 15m 00s = 1 minute = 60 seconds

Total Difference = 1 minute 38 seconds (= 38s + 60s = 98 seconds)

Alternative total seconds method check:

Target = (2 × 3600) + (15 × 60) = 7200 + 900 = 8,100 seconds

Bradley = (2 × 3600) + (13 × 60) + 22 = 7200 + 780 + 22 = 8,002 seconds

8,100 − 8,002 = 98 seconds = 1 minute 38 seconds (verified)

Final Answer with correct time units:

[A1 answer]

Bradley was 1 minute 38 seconds faster than target (98 seconds)

(Total for Question 3 is 5 marks)

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SECTION B: The house

Answer all questions in this section.

Write your answers in the spaces provided.

4 Matt and Julia find a two bedroom house to rent.
Matt will have the bigger bedroom.

They have this information about the costs they need to pay between them before they can move in.

- Rent for 1 month £895
- Administration fee 0.7 × monthly rent
- Deposit 1.25 × monthly rent

Matt and Julia will split the total of these costs in the ratio 3 : 2

(a) How much does Matt have to pay before they move in?

(6)

Use the box below to show clearly how you get your answer.

Step 1: Calculate individual moving-in costs based on £895 monthly rent *[M1 items]*

- 1 month rent = £895.00
- Admin fee = 0.7 × £895 = £626.50
- Deposit = 1.25 × £895 = £1,118.75

Step 2: Calculate total cost to be paid before moving in *[M1 total]*

Total Cost = £895.00 + £626.50 + £1,118.75

Total Cost = £2,640.25

Step 3: Split total cost in ratio Matt : Julia = 3 : 2 *[M1 ratio]*

Total ratio parts = 3 + 2 = 5 parts

Value of 1 part = £2,640.25 ÷ 5 = £528.05

Step 4: Calculate Matt's share (3 parts) *[M1 Matt share]*

Matt pays = 3 × £528.05 = £1,584.15 *[A1 answer]*

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Verification Breakdown for Moving-In Costs:

- Matt's 3 parts = 3 × £528.05 = £1,584.15
- Julia's 2 parts = 2 × £528.05 = £1,056.10
- Total Check = £1,584.15 + £1,056.10 = £2,640.25 (matches total cost exactly)

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Matt wants to buy some furniture for the house.
He borrows £3000 to pay for the furniture.
He needs to pay a borrowing fee.

Matt needs to repay the £3000 and the borrowing fee over 18 months.

He uses this formula to work out the amount of money he will repay each month.

$$R = \frac{A \times 1.034}{n}$$

R is the amount to repay each month (£).
A is the amount borrowed (£).
n is the number of months to pay the money back.

Matt thinks the amount he will need to repay each month is more than £170

(b) Will he need to repay more than £170 each month?
Show a check of your working. **(4)**

Use the box below to show clearly how you get your answer.

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Step 1: Substitute loan amount (£3000) and 18 months into formula [M1 formula]

Formula: $R = (A \times 1.034) \div n$

Total to repay $(A \times 1.034) = £3{,}000 \times 1.034 = £3{,}102.00$

Step 2: Calculate monthly repayment (R = Total ÷ 18 months) [M1 monthly]

Monthly Repayment $R = £3{,}102.00 \div 18$

$= £172.333\ldots \approx £172.33$ per month

Step 3: Compare with Matt's claim (> £170 per month)

£172.33 > £170.00 (exceeds £170 by £2.33 per month)

[A1 decision]

Conclusion: Yes, Matt will need to repay more than £170 each month.

Use the box below to show your check.

☑

Reverse calculation check: [B1 check]

18 × £172.333... = £3,102.00; £3,102 ÷ 1.034 = £3,000.00 (matches loan exactly)

Alternative: 18 × £170 = £3,060 < £3,102 (confirms repayment > £170) (correct)

(Total for Question 4 is 10 marks)

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5 The sketch shows the dimensions of the lounge floor.
All the corners in the room are right angles.

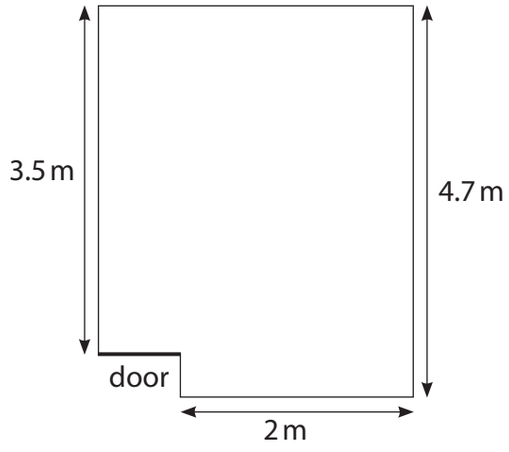


Diagram **not** accurately drawn

The door is 1.25 m wide.

Julia wants to put string lighting in the lounge.
She wants the lights to go all the way around the room.
The lights will go above the height of the door.

Julia wants enough string lighting to cover at least $1\frac{1}{2}$ times the total distance around the room.

She thinks 25 metres of string lighting will be enough.

(a) Is Julia correct?
Show why you think this. **(4)**

Use the box below to show clearly how you get your answer.

Step 1: Calculate total perimeter around lounge room floor *[M1 perimeter]*

Width = 2.0 m + 1.25 m (door) = 3.25 m, Length = 4.7 m

Perimeter = 2 × (4.7 + 3.25) = 2 × 7.95 = 15.9 m

Step 2: Calculate lighting required (1.5 times distance around room) *[M1 multiplier]*

Required Length = 1.5 × 15.9 m = 23.85 m

23.85 m ≤ 25 m (25 metres provides 1.15 m extra lighting)

[A1 decision]

Conclusion: Yes, Julia is correct (25 metres will be enough).

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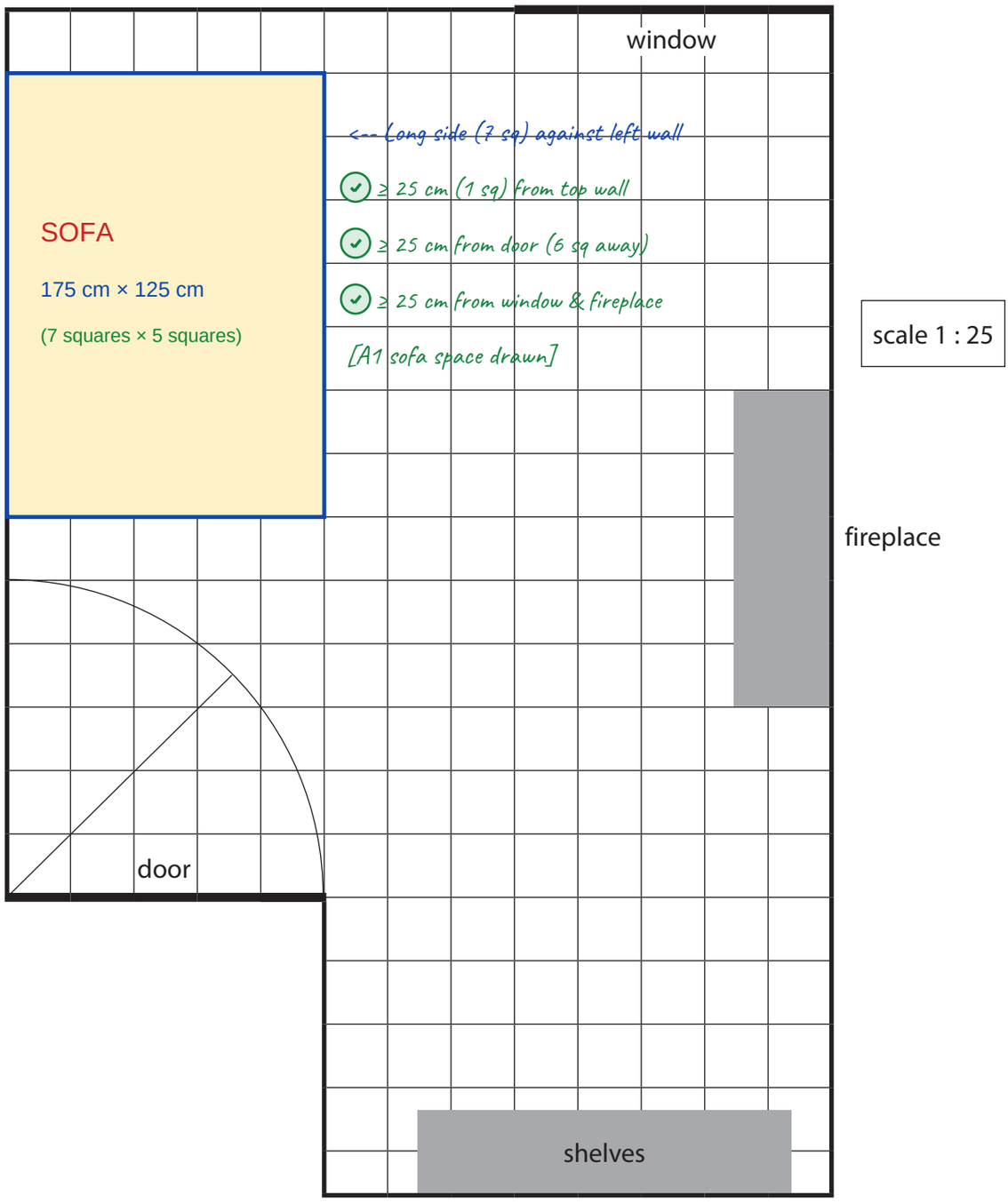
Julia has a large sofa that she wants to put in the lounge.
The sofa needs a rectangular space 175 cm by 125 cm.

Julia wants the space for the sofa to have

- the longest side against a wall
- at least 25 cm from the door, the window, the shelves and the fireplace.

Julia has an accurate drawing of the floor of the lounge.

(b) Draw the space for the sofa on the grid below.
Remember to use the scale given. **(2)**



(Total for Question 5 is 6 marks)

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SECTION C: Limousine hire

Answer all questions in this section.

Write your answers in the spaces provided.

6 Ellie is the manager of a limousine hire company.

This is a plan of the driveway.

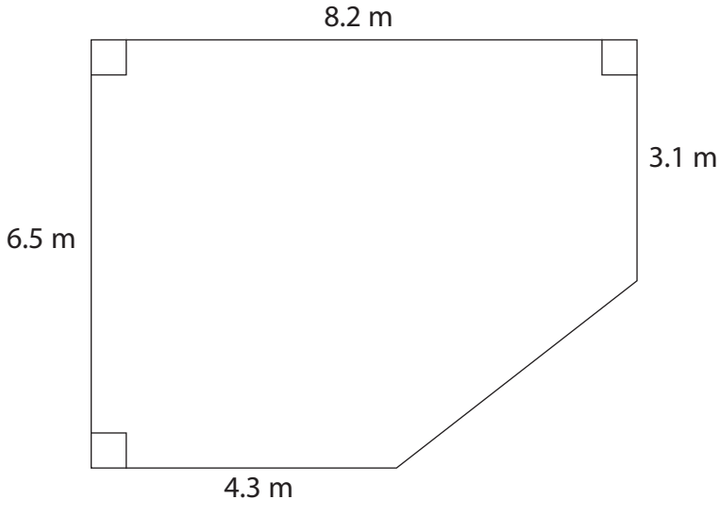


Diagram **not** accurately drawn

Ellie wants to cover the driveway with paving stones.
The paving stones can be cut to fit the space.

Ellie will buy the paving stones in packs.
Each pack covers an area of 4.51 m^2

What is the least number of packs of paving stones Ellie needs to buy?

(5)

Use the box below to show clearly how you get your answer.

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Step 1: Calculate total floor area of limousine floor [M1 area]

Method: Outer Bounding Rectangle minus cut-out corner triangle

Outer dimensions = 8.2 m long by 6.5 m wide

Outer Rectangle Area = 8.2 × 6.5 = 53.30 m^2

Step 2: Find cut-out dimensions & subtract cut-out area [M1 cutout]

Missing triangle base = 8.2 − 4.3 = 3.9 m; height = 6.5 − 3.1 = 3.4 m

Cut-out Triangle Area = 0.5 × 3.9 × 3.4 = 6.63 m^2

Total Floor Area = 53.30 − 6.63 = 46.67 m^2

Step 3: Calculate number of packs required (each pack covers 4.51 m^2) [M1 packs]

Number of packs = 46.67 m^2 ÷ 4.51 m^2 per pack

= 10.348... packs

Must round UP to whole packs since tiles cannot be bought partially

Packs of flooring required = 11 packs [A1 answer]

(Total for Question 6 is 5 marks)

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7 Adnan, Fayez and Toni are the 3 drivers working on Saturday.

Ellie needs to complete a booking sheet for each driver.
Some bookings have already been made for Saturday.

Ellie needs to add these new bookings to the booking sheets.

Client	**Time**
Mr Harding	9 am to 12 pm
Barker family	9.30 am to 11 am
Mr and Mrs Khan	12 pm to 1.30 pm
Ms Singh	1 pm to 2.30 pm

Ellie needs to allow at least 45 minutes between each booking.

Complete the booking sheets for the drivers.

(3)

Use the booking sheets on the next page.

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Driver: Adnan	**Day:** Saturday
Time	**Client**
9 am to 12 pm	Magenty family
1 pm to 2.30 pm	Ms Singh
✓ Gap after Magenty: 12.00 pm to 1.00 pm = 60 mins (≥ 45 mins)	

Driver: Fayez	**Day:** Saturday
Time	**Client**
9 am to 12 pm	Mr Harding
1 pm to 3 pm	Mr and Dr Pacitti
✓ Gap before Pacitti: 12.00 pm to 1.00 pm = 60 mins (≥ 45 mins)	

Driver: Toni	**Day:** Saturday
Time	**Client**
9.30 am to 11 am	Barker family
12 pm to 1.30 pm	Mr and Mrs Khan
2.30 pm to 5 pm	Lee family
	[A1 full schedule]

(Total for Question 7 is 3 marks)

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8 At the beginning of the week Fayez writes down the mileage of his limousine.

Mileage
2529

At the end of the week the mileage is 2834
Fayez knows he used 75.2 litres of petrol during this week.

The handbook states that the limousine uses 1 gallon of petrol for every 20 miles.

Fayez wants to know if his limousine used 1 gallon of petrol for every 20 miles that week.

1 gallon = 4.54 litres

Did his limousine use 1 gallon of petrol for every 20 miles that week?

(4)

Use the box below to show clearly how you get your answer.

Step 1: Calculate total miles driven that week *[M1 miles]*

Distance = 2,834 (end) − 2,529 (start) = 305 miles

Step 2: Convert litres of petrol used into gallons (1 gallon = 4.54 litres) *[M1 gallons]*

Gallons = 75.2 litres ÷ 4.54 litres/gal = 16.56387... gallons

Step 3: Calculate actual fuel economy in miles per gallon (mpg) *[M1 mpg]*

Fuel Economy = 305 miles ÷ 16.56387... gallons

= 18.413... ≈ 18.4 mpg

Step 4: Compare with handbook specification (20 mpg)

18.4 mpg < 20 mpg (achieved 1.6 mpg less than specified)

[A1 decision]

Conclusion: No, his limousine did NOT use 1 gallon for every 20 miles.

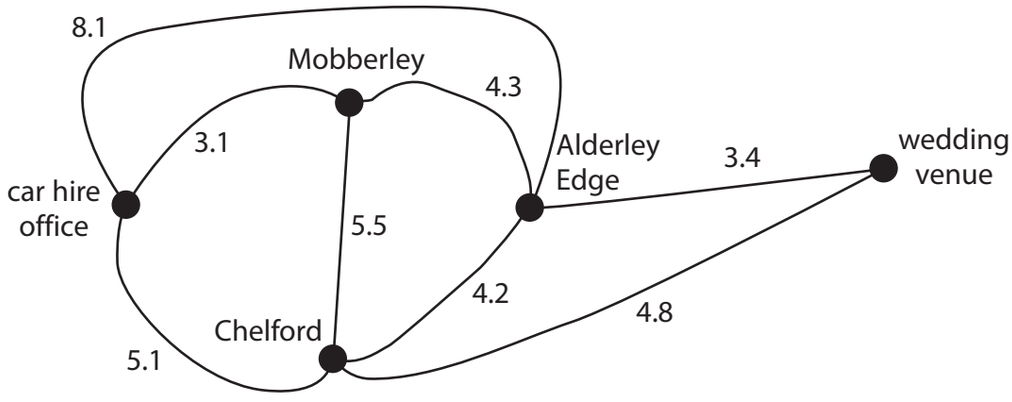
(Total for Question 8 is 4 marks)

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9 On Friday Toni will take some clients to a wedding venue.

He will
- start at the car hire office
- pick up clients in Alderley Edge, Chelford and Mobberley
- then take all of these clients to the wedding venue.

The plan shows distances in miles.



Toni wants to use the shortest route.

(a) Find the shortest route for Toni.
Work out the distance of this route. **(3)**

Use the box below to show clearly how you get your answer.

Route 1: Office --> Mobberley --> Chelford --> Alderley --> Wedding

Distance = 3.1 + 5.5 + 4.2 + 3.4 = 16.2 miles (SHORTEST ROUTE)

Route 2: O --> M --> A --> C --> W = 3.1 + 4.3 + 4.2 + 4.8 = 16.4 miles

Route 3: O --> C --> M --> A --> W = 5.1 + 5.5 + 4.3 + 3.4 = 18.3 miles

Shortest Distance = 16.2 miles [A1 distance]

(b) What else should Toni take into account when he plans his route?
(1)

Write your answer in the box below.

Traffic congestion and rush hour delays on Friday afternoon. [A1 factor]

(Alternative valid factors: Road works, road closures, or speed limits).

(Total for Question 9 is 4 marks)

TOTAL FOR PAPER IS 48 MARKS

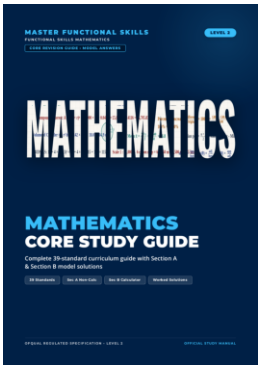
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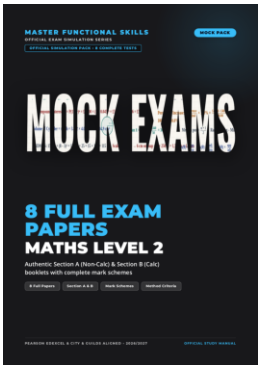


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